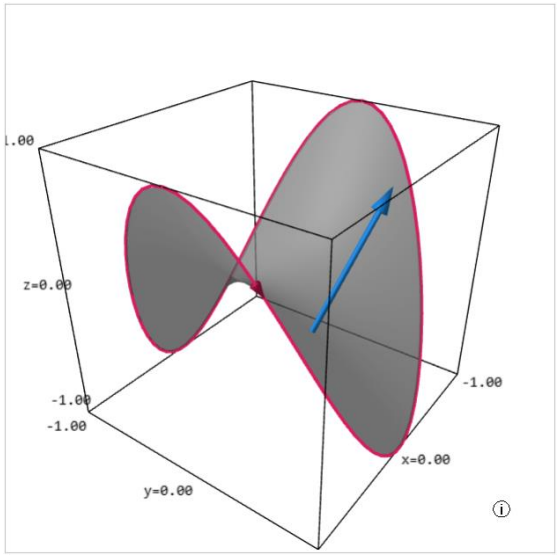
Vector Calculus and Complex Analysis Classwork (2025/3/24)

Name:

Student ID:

Consider the surface $S=\{(x,y,x^2-y^2): x^2+y^2\le 1\}$. Verify Stokes's Theorem for $\boldsymbol{F}(x,y,z)=\langle z-x, y+z, xy\rangle$.



Solution

(a) We first calculate the circulation of $\mathbf{F} = \langle z - x, y + z, xy \rangle$ around the curve C with parameterization given by $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t) \rangle$ with $0 \leq t \leq 2\pi$. Note here that $\mathbf{r}'(t) = \langle -\sin(t), \cos(t), -2\sin(2t) \rangle$. Applying Theorem 12.3.6 to calculate the circulation, we have

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^{2\pi} \langle \cos(2t) - \cos(t), \sin(t) + \cos(2t), \cos(t)\sin(t) \rangle \\ &\qquad \cdot \langle -\sin(t), \cos(t), -2\sin(2t) \rangle \, dt \\ &= \int_0^{2\pi} 2\cos(t)\sin(t)\, dt \\ &\quad - \int_0^{2\pi} \cos(2t)\sin(t)\, dt \\ &\quad + \int_0^{2\pi} \cos(2t)\cos(t)\, dt \\ &\quad - \int_0^{2\pi} 2\cos(t)\sin(t)\sin(2t)\, dt \\ &= 0 - 0 + 0 - \pi = -\pi\end{aligned}$$

$\text{curl}(\boldsymbol{F}) = \langle x-1, 1-y, 0 \rangle.$

$S = \{(x, y, x^2 - y^2) : x^2 + y^2 \leq 1\}.$

$r_x = (1, 0, 2x)$

$r_y = (0, 1, -2y)$

$r_x \times r_y = (-2x, 2y, 1)$

$\iint_S \langle x-1, 1-y, 0 \rangle \langle -2x, 2y, 1 \rangle \, dxdy$

$= \iint_S -2x(x-1) - 2y(y-1) dxdy$

$= -2\int_0^{2\pi} \int_0^1 \left(r\cos\theta(r\cos\theta - 1) + r\sin\theta(r\sin\theta - 1) \right) r dr d\theta$

$= -2\int_0^1 \int_0^{2\pi} \left(r^3\cos^2\theta - r^2\cos\theta + r^3\sin^2\theta - r^2\sin\theta \right) d\theta dr$

$= -2\int_0^1 2\pi r^3 dr$

$= -\pi$